

Each manufacturer has its own strategy on what to test and where

Everybody uses mobile devices, but how many of us know the secrets of testing these? Lauri Antila, vice president-marketing and business development, JOT Automation spoke to EFY about the latest in mobile testing and the role of automation. JOT Automation, founded in 1988, develops test and production solutions for smart products, components and electronics



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Q: What are the most important parameters that should be tested in a mobile device?

The job starts from the board-level testing, i.e., testing the phones as PCBs and PCB-panels. In addition to electrical testing of the device, we should also take care of software flashing and RF-tuning. In other words, completely integrated and customised test solutions are needed. In the final testing phase, we simulate the use of smart phones using a variety of advanced sensor and robot technologies in a repeatable and reliable environment. In practice, our solutions test the proper functionality of microphones, loudspeakers, cameras, radios, dimensions, push buttons, signal lights and displays, touchscreens, and also the testing of all the physical connectors and interfaces, such as USB and audio plug.

Q: What would be an optimal method to test mobile devices, that allows minimum time to market?

The best solution would be one that is suitable for a variety of test strategies, processes, end products and applications, which makes them future-proof investments for manufacturers of smart devices. This ensures cost effectiveness for the test equipment. One of the key benefits of such a solution is that the very same test solution can be taken from R&D to production to secure efficient and swift production ramp up. The other key benefit of the solution is the modular system architecture, which enables it to easily adapt to production-volume variations during a product's life cycle, and also between the production lines and factories.

Q: How is the industry speeding up development cycles and overcoming capacity challenges?

Modularity and re-configurability is a must here. It is simply not possible to design manufacturing solutions from the scratch for new products. The answer for these kind of needs is platform(s) that can easily be adopted to new product variant or even totally new product. In practice you need only new test fixtures when changing product in production. This can be achieved in production testing both at board level and product level testing. Once verified and integrated to a customer's production testing processes, changing the products to be tested is rather simple. Even test fixtures are based on

standard components, so basically ramping up new products is about configuring new fixtures.

Q: How important is automation in production testing?

As a simple example, you would use automated solution, instead of operator's eye, to verify the functionality of the camera zoom. And to get exact automatically generated report about the quality of the process for each step and functionality. In general, this will put OEMs to better position.

Q: Are the test systems developing the way the industries want them to?

Smart phones get smarter and smarter and require even more testing than before. To keep the costs of testing on an acceptable level it is very important to have all the parties involved in development work. Many of the decisions regarding the efficiency of production testing are already made by chipset providers.

Q: What flaws are caused during manufacturing process?

During board assembly you can have problems with components or with the assembly process of components. In product assembly you don't necessarily get smart phone screen exactly placed where you want to. Or maybe the screen wasn't glued perfectly because of some dirt. Did we download the latest software release approved? A lot of things go wrong, and the sooner you detect the flaw the better. Product design and its quality is one thing, quality of the incoming components is another thing and quality of the manufacturing process as such is still another thing. However, everything is related and each manufacturer has its own strategy on what to test and where.

Q: Which features on a phone are the most challenging to test?

Many electromechanical features can be tricky. One example could be the exact functionality of push buttons. Human fingers are very sensitive but every finger is different. Automating that touch sensitiveness is pretty challenging. Another example of the challenging testing is the testing of antennae in production environment. ●